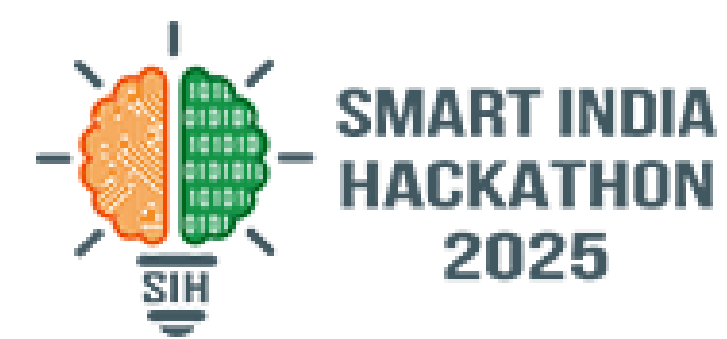
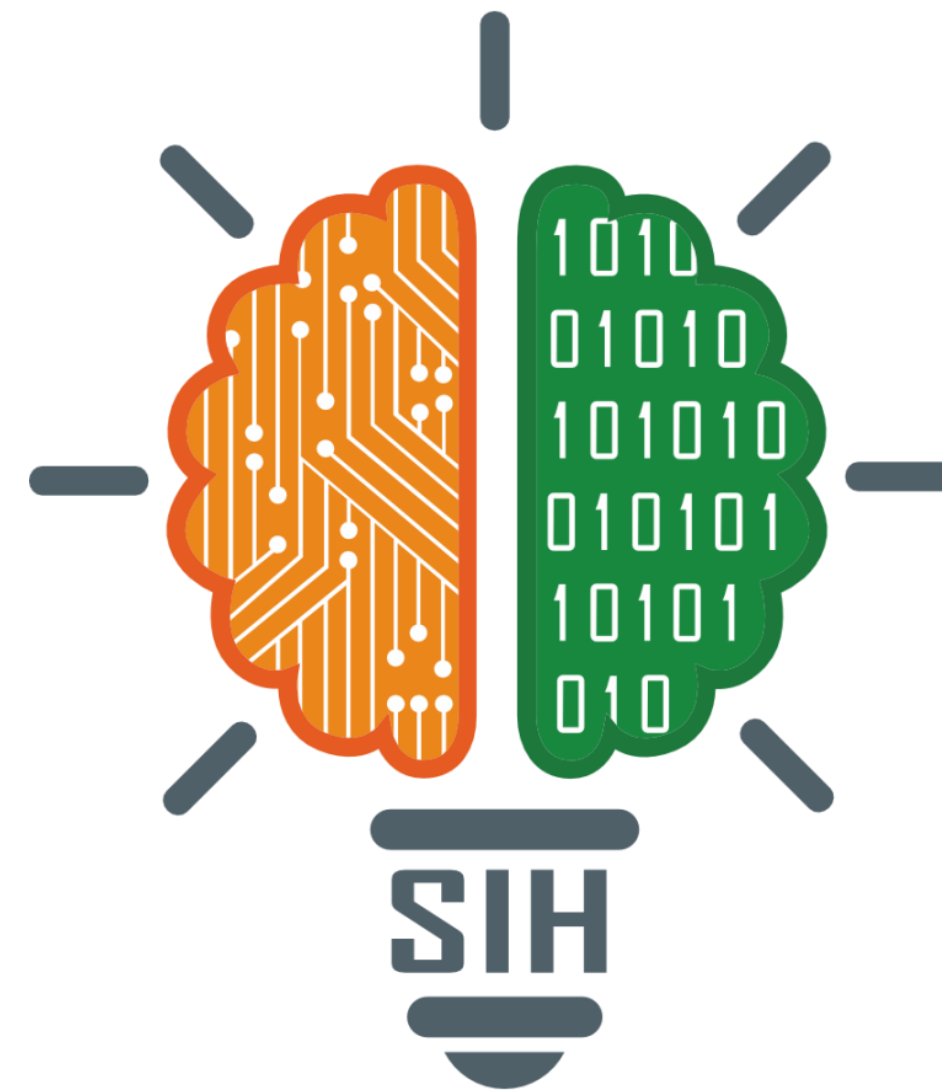


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID** – SIH26063
- **Problem Statement Title-** Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 138601
- **Team Name -** NIRVANA



Problem:

- **Fragmented Three-Region Archives:** Decades of expedition reports, datasets, 4K media, and historical logs are scattered across separate silos for Antarctica, the Arctic, and the Himalayas.
- **Complex Cryospheric Science:** Specialized glaciological, atmospheric, and oceanographic research is difficult for students, educators, and the public to discover and comprehend.
- **Manual Outreach Preparation:** Turning raw expedition data into bilingual (English + Hindi) web articles and press releases requires hours of repetitive manual drafting with risk of factual errors.

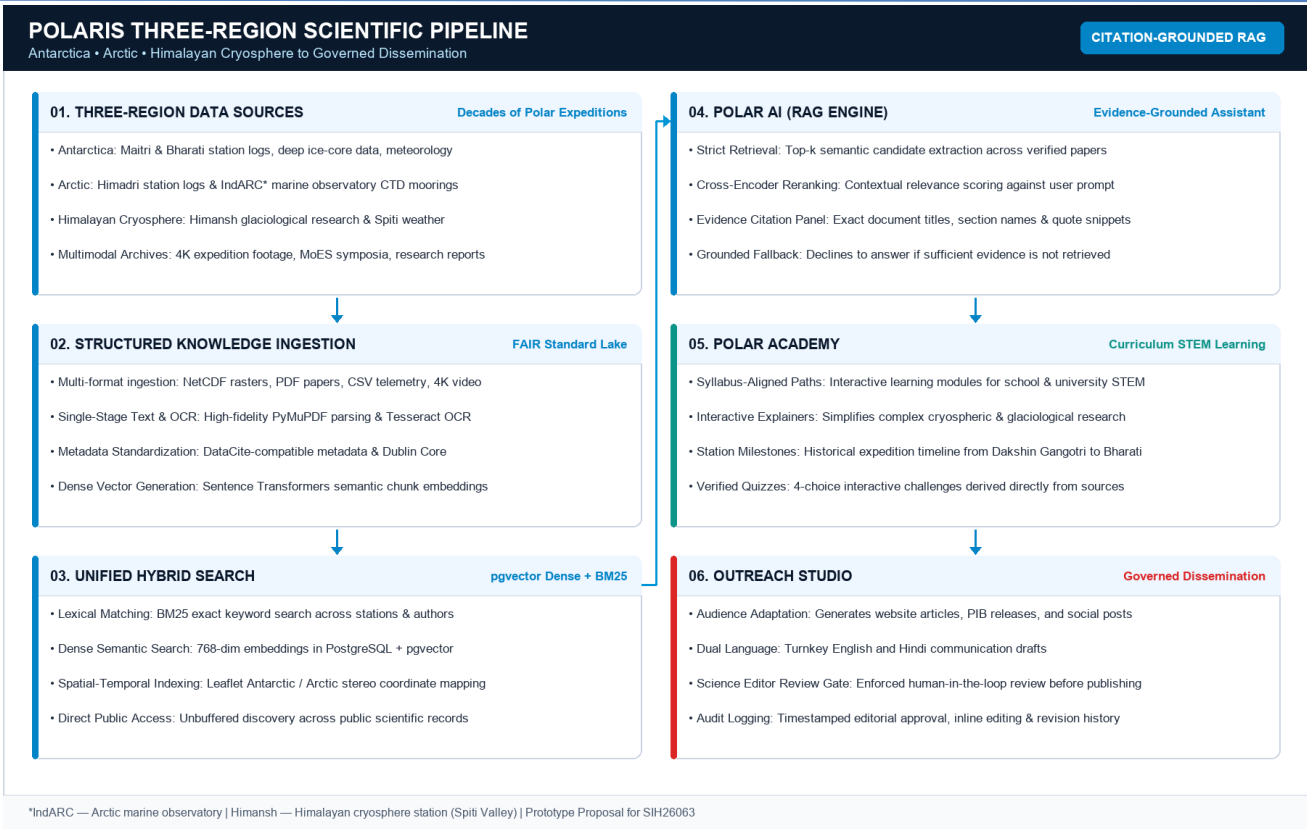
40+ Years
Expeditions
Decades of Polar Science

3 Regions
Stations & Obs
Maitri, Bharati, Himadri, IndARC*, Himansh

Citation RAG
Evidence-Based
Source & Page Citations

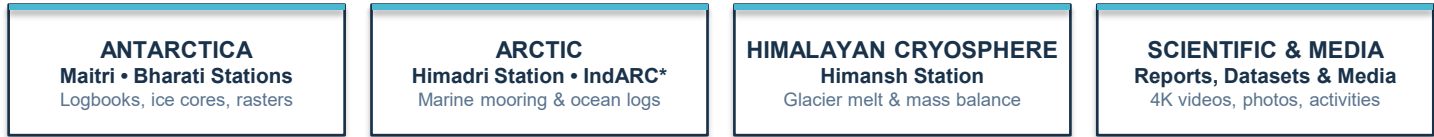
How It Addresses the Problem:

- **Unified Three-Region Knowledge Lake:** Consolidates expedition reports, datasets, 4K drone media, satellite rasters, and institutional activities across Antarctica, Arctic, and the Himalayas.
- **Citation-Grounded RAG Assistant:** POLAR AI answers science queries with retrieved evidence, exact document titles, section headers, and verbatim quotes; declines when evidence is missing.
- **Interactive 3-Region Station GIS:** Geospatial exploration of Maitri, Bharati, Himadri, IndARC*, and Himansh with interactive layers, cruise tracks, and environmental demo feeds.
- **Multimodal Media & Institutional Archive:** Dedicated catalog for 4K expedition photography, research videos, MoES flag-offs, symposia, and historical milestone chronicles.
- **Governed AI Outreach Studio:** Generates bilingual (English + Hindi) articles, PIB press releases, and student quizzes backed by a mandatory Science Editor review gate.

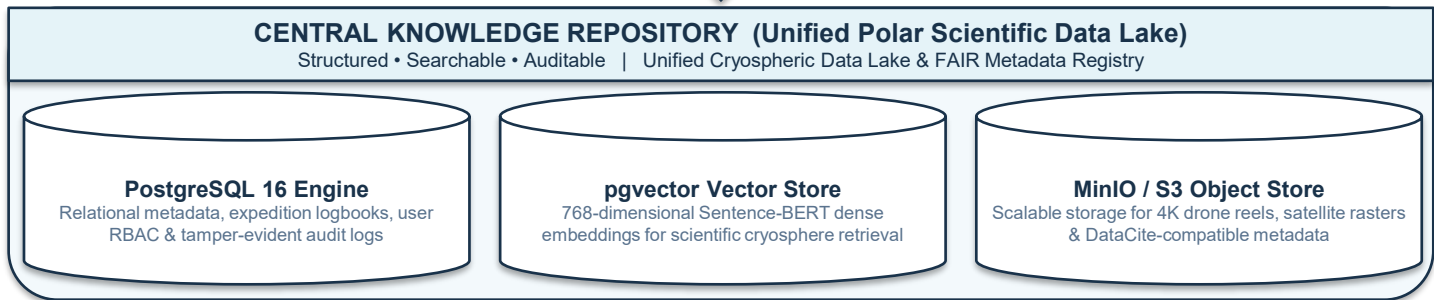
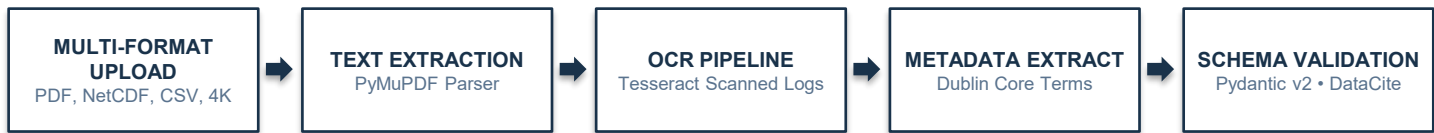


SYSTEM ARCHITECTURE & INFORMATION FLOW

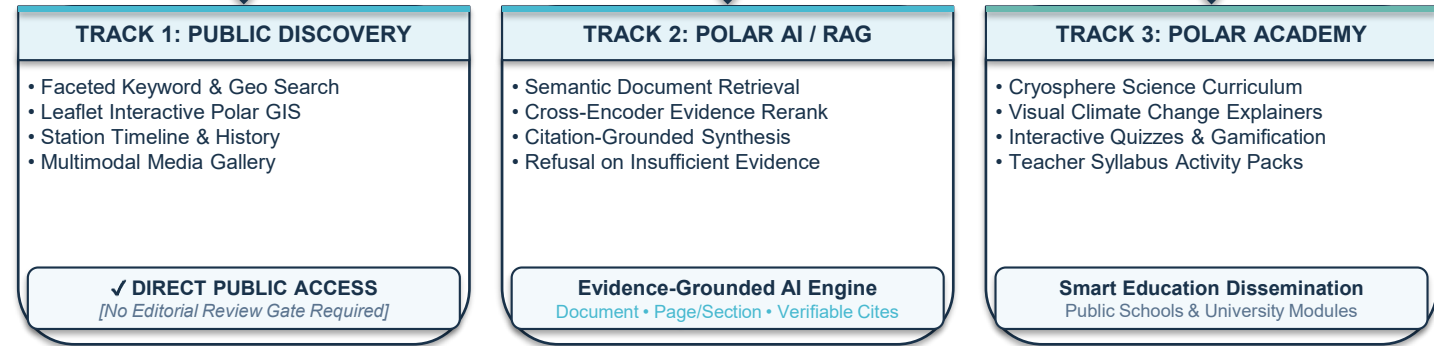
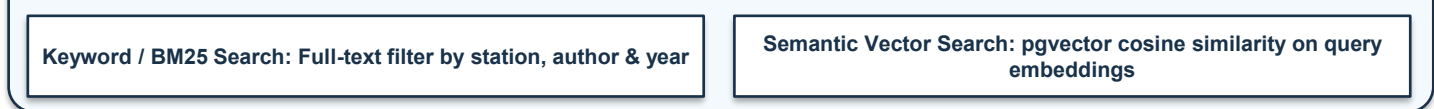
1. POLAR DATA SOURCES (Three-Region Scientific Archives & In-Situ Feeds)



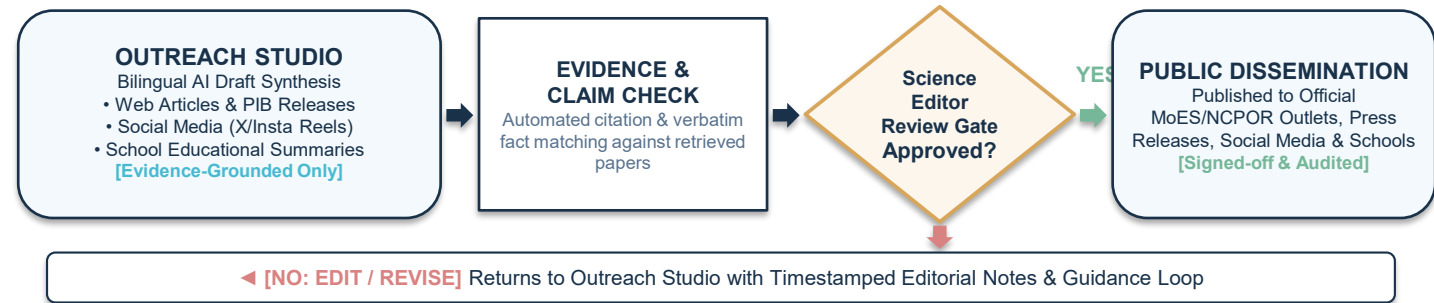
2. RESEARCHER & ADMIN INGESTION PIPELINE (Authorized Ingestion & Validation)



4. HYBRID SEARCH ENGINE (Dual-Mode Lexical & Semantic Retrieval)



5. GOVERNED OUTREACH STUDIO & HUMAN SCIENCE EDITORIAL REVIEW GATE



Technology Stack	Why this choice & Role
Frontend	
Next.js + TypeScript	Modern, type-safe, server-rendered scientific web portal
Tailwind CSS + shadcn/ui	Accessible, responsive polar theme: deep navy, ice blue, cyan
Leaflet + OpenStreetMap	Interactive polar GIS station explorer & geospatial tracking
Backend	
Python + FastAPI	High-concurrency async REST API for RAG, search & data ingestion
Pydantic v2	Strict schema validation for DataCite/Dublin Core metadata models
REST APIs	Decoupled, modular communication between UI, RAG & data services
Uvicorn ASGI	Production-grade asynchronous web server with worker management
AI & Knowledge Retrieval	
PyMuPDF + Tesseract	High-accuracy extraction from scanned expedition PDFs & field logs
Sentence Transformers	High-dimensional scientific embeddings for polar research corpus
POLAR AI RAG Engine	Citation-grounded question answering with strict evidence verification
Outreach Generator	Multi-platform content synthesis: Web, Instagram, LinkedIn, Academy
Data & Security	
PostgreSQL + pgvector	Unified relational storage & high-performance vector similarity search
JWT + PBKDF2 / Bcrypt	Secure authentication with granular RBAC: Public, Researcher, Editor, Admin
Role-Based Access (RBAC)	Granular permissions protecting unreleased datasets & editorial workflows
Audit Logging	Tamper-evident logs of all content edits, approvals & dataset downloads
Storage & Cloud Deployment	
MinIO / S3 Storage	Scalable object store for expedition high-res imagery, video logs & raw datasets
Recharts Analytics	Interactive visualization of polar dataset trends, publications & outreach metrics
Docker & Compose	Containerized, reproducible one-command deployment across cloud & on-premise
Git CI/CD	Automated testing, version control, and multi-stage container builds

POLARIS System Interface & Live Modules



Fig 1: POLARIS Interactive Station GIS & Simulated Telemetry [DEMO DATA]

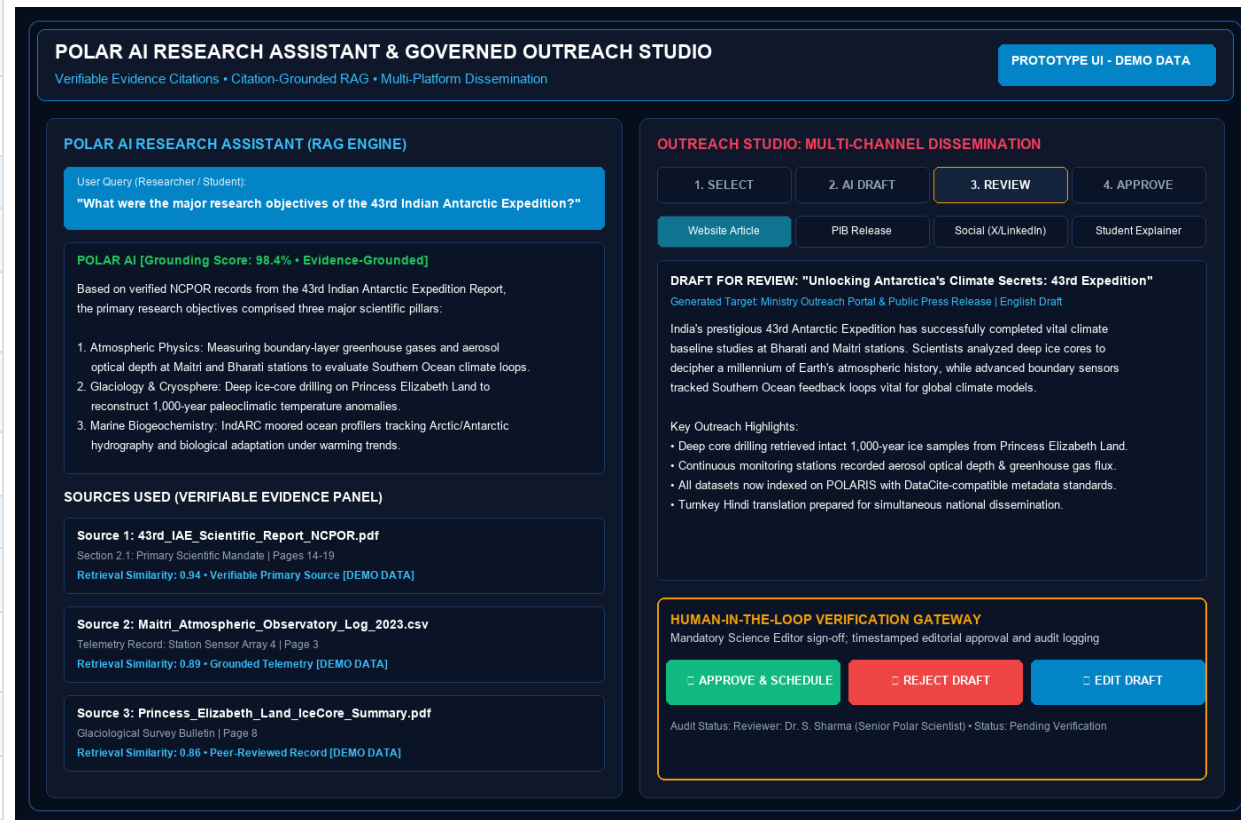


Fig 2: POLARIS Citation-Grounded RAG Assistant & Outreach Studio [PROTOTYPE UI - DEMO DATA]

FEASIBILITY

TECHNICAL

- Prototype Architecture: Built on FastAPI, Next.js, PostgreSQL/pgvector & Leaflet. Designed for integration with authorized institutional archives.
- Three-Region Sensor Support: Standardized ingestion of NetCDF telemetry, satellite rasters, 4K media, and scanned historical logs.
- Decoupled Microservices: Independent horizontal scaling of RAG retrieval, OCR workers, and media streaming services.

OPERATIONAL

- Editorial Quality Gate: Mandatory Human-in-the-Loop review ensures factual accuracy before external publication.
- Bilingual Dissemination: Turnkey generation of English and Hindi outreach materials, reducing repetitive administrative drafting effort.
- User-Friendly Interface: Intuitive self-service deposit UI requires no technical training for returning expedition scientists.

LEGAL & SECURITY

- Privacy & Compliance: Designed with privacy and DPDP compliance considerations & Antarctic Treaty data guidelines.
- Enterprise Security: JWT tokens, PBKDF2 hashing, granular role-based RBAC, and automated embargo scheduling.
- Audit Logging: Timestamped audit logs record all uploads, RAG queries, inline edits, and editorial approvals.

FINANCIAL

- Open-Source-First: Open-source-first technology stack (PostgreSQL, FastAPI, Next.js, Python) avoids proprietary license fees.
- Flexible Infrastructure: Containerized deployment using Docker runs seamlessly on existing on-prem servers or cloud.

VIABILITY

SCALABLE + SUSTAINABLE

- Scalable Knowledge Lake: Enterprise PostgreSQL + pgvector cluster with MinIO S3 object storage handles multi-terabyte data.
- Long-Term Sustainability: Modular architecture supports future satellite constellations, IoT moorings, and UAV sensor arrays.

EXTENSIBLE + DEPLOYABLE

- Standardized Interoperability: Native REST APIs link directly to SCAR, WMO Global Cryosphere Watch, and DataCite-compatible registries.
- Containerized Deployment: Containerized deployment using Docker ensures reproducible, environment-independent setups.

TRUST-READY

- Verifiable Citations: Polar AI strictly cites document titles, page numbers & verbatim quotes; declines if ungrounded.
- Institutional Oversight: Multi-tier editorial sign-off ensures public-facing press releases meet MoES scientific standards.

COST-EFFECTIVE

- Workflow Efficiency: Reduces repetitive multi-channel outreach drafting effort, freeing science communication officers for strategic institutional engagements.
- Zero Licensing Overhead: Built entirely on open-source production foundations (PostgreSQL, FastAPI, Next.js), eliminating expensive proprietary enterprise software fees.
- Sustainable Operational Footprint: Lightweight asynchronous architecture runs reliably on commodity cloud or existing institutional server infrastructure.

POTENTIAL CHALLENGES & MITIGATIONS

1. DATA HETEROGENEITY & QUALITY

RISK: Decades of heterogeneous records across papers, logs, CSVs, and 4K media.
STRATEGY: Automated PyMuPDF OCR + CLIP multimodal embeddings + strict DataCite 4.4 validation.

2. AI HALLUCINATION & ACCURACY

RISK: AI hallucinating incorrect polar station coordinates or unverified conclusions.
STRATEGY: Dense pgvector retrieval + mandatory verbatim quotes + explicit refusal fallback when ungrounded.

3. MULTIMODAL MEDIA VOLUME

RISK: Storage & bandwidth bottlenecks from 4K drone footage, satellite rasters, and video logs.
STRATEGY: MinIO / S3 chunked object storage + optimized CDN delivery + automated multi-resolution streaming.

4. UNAPPROVED CONTENT DISSEMINATION

RISK: Unverified AI drafts accidentally published to public channels, impacting credibility.
STRATEGY: Enforced Human-in-the-Loop review gate; initial state strictly locked as Draft until authorized sign-off.

 DIRECT IMPACT ON TARGET USERS



RESEARCHERS: DISCOVER

- Unified 3-Region Discovery: Centralized search across decades of expedition reports, NetCDF datasets, and station logs across Antarctica, Arctic, and the Himalayas in one query.
- Self-Service Researcher CMS: Standardized deposit portal with DataCite-compatible metadata and embargo management accelerates new discoveries.



STUDENTS: LEARN

- STEM Polar Academy: Interactive timelines, simplified climate explainers, and gamified quizzes generated directly from verified peer-reviewed polar science.
- Career Inspiration: Inspires next-generation STEM careers in glaciology, oceanography, polar physics, and Himalayan cryosphere research.



EDUCATORS: TEACH

- Verifiable Classroom Resources: Curated syllabus-aligned learning paths, ready-to-teach polar milestones, and direct primary literature citations.
- Source-Linked Learning: Eliminates hours spent cross-referencing informal, unreliable internet claims with 1-click evidence links.



OUTREACH TEAMS: COMMUNICATE

- Bilingual Multi-Channel AI: Automated generation of website articles, PIB press releases, X threads, and student explainers in both English and Hindi.
- Workflow Efficiency: Reduces repetitive multi-platform content preparation while enforcing mandatory human-in-the-loop editorial approval.

 BENEFITS OF THE SOLUTION



THREE-REGION KNOWLEDGE SCOPE

- Three-Region Knowledge Lake: Unifies fragmented expedition archives, research papers, and environmental datasets across Antarctica, Arctic, and the Himalayan Third Pole.
- FAIR Preservation: Preserves India's rich polar scientific heritage and guarantees FAIR-compliant accessibility for the scientific community.



PUBLIC: UNDERSTAND

- Expanding Polar Access: Expands access to polar science for students and learners, translating complex cryospheric and sea-ice data into engaging stories.
- Strategic Awareness: Fosters public awareness of India's strategic research stations & observatories (Maitri, Bharati, Himadri, IndARC*, Himansh).



OPERATIONAL EFFICIENCY

- Inter-Agency Workflow: Eliminates redundant data requests and manual cross-referencing between research divisions and MoES communication cells.
- Faster Discovery: Enables faster discovery and retrieval of relevant information across decades of archived records.



EVIDENCE-GROUNDED AI

- Verifiable Citations: Retrieved sources and citations help users verify AI-generated responses with verbatim quotes and exact page references.
- Evidence Check: Can decline to answer when sufficient supporting evidence is not retrieved, safeguarding institutional credibility.

01. SIH26063 Problem Statement — Ministry of Earth Sciences (MoES) / NCPOR (2026)

Role: Official government mandate for integrated polar science outreach & knowledge repository.
Verified Link: <https://sih.gov.in> | <https://moes.gov.in>

02. NCPOR National Polar Data Center & Cryospheric Archives — MoES / NCPOR (2024)

Role: Baseline metadata & telemetry records for Antarctica (Maitri, Bharati), Arctic (Himadri, IndARC*), Himalaya (Himansh).
Verified Link: <https://data.ncpor.res.in> | <https://ncpor.res.in>

03. Retrieval-Augmented Generation for Knowledge-Intensive NLP — Lewis et al. (NeurIPS 2020)

Role: Foundational architecture grounding POLAR AI answers strictly in retrieved repository evidence.
Verified Link: <https://arxiv.org/abs/2005.11401> | <https://doi.org/10.48550/arXiv.2005.11401>

04. DataCite Metadata Schema 4.4 for Research Datasets — DataCite Working Group (2021)

Role: Standardized schema for research datasets, expedition reports, DOIs, and creator relationships.
Verified Link: <https://schema.datacite.org/meta/kernel-4.4/> | <https://datacite.org>

05. Dublin Core Metadata Terms (DCMI) — ISO Standard 15836-1:2017 (2017)

Role: Multidisciplinary metadata taxonomy for polar publications, media archives & institutional activities.
Verified Link: <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/> | <https://iso.org>

06. CLIP: Learning Transferable Visual Models — Radford et al. (ICML 2021)

Role: Natural-language cross-modal semantic discovery across polar photography, satellite rasters & drone videos.
Verified Link: <https://arxiv.org/abs/2103.00020> | <https://github.com/openai/CLIP>

07. Sentence-BERT: Sentence Embeddings using Siamese Networks — Reimers & Gurevych (EMNLP 2019)

Role: Powers fast, dense vector indexing and hybrid semantic retrieval across polar research corpus.
Verified Link: <https://arxiv.org/abs/1908.10084> | <https://www.sbert.net>

08. WMO Global Cryosphere Watch (GCW) Data Standards & Formats — WMO (2023)

Role: Cryospheric observation standards, sea-ice data formatting & polar weather telemetry integration.
Verified Link: <https://globalcryospherewatch.org> | <https://wmo.int>

09. Leaflet & EPSG:3031 / EPSG:3995 Polar Projections — EPSG / OpenStreetMap (2023)

Role: Interactive geospatial rendering of polar research stations, expedition cruise tracks & sensor sites.
Verified Link: <https://leafletjs.com> | <https://epsg.io/3031>

10. FastAPI Asynchronous Framework & pgvector PostgreSQL Extension — Tiangolo (2024)

Role: Ultra-low-latency vector similarity search and scalable asynchronous REST API microservices.
Verified Link: <https://fastapi.tiangolo.com> | <https://github.com/pgvector/pgvector>

TECHNICAL STANDARDS & ARCHITECTURAL ALIGNMENT MATRIX

- **Institutional Mandate & Polar Data (Refs [01, 02]):** MoES / NCPOR SIH26063 guidelines, Three-Region coverage (Antarctica, Arctic, Himalayas), and national cryosphere repository compatibility.
- **FAIR Metadata & Data Interoperability (Refs [04, 05]):** DataCite 4.4 kernel and Dublin Core (ISO 15836-1:2017) schemas ensuring Findable, Accessible, Interoperable, and Reusable scientific records.
- **AI, Semantic Search & Multimodal Indexing (Refs [03, 06, 07]):** Retrieval-Augmented Generation (Lewis et al.), Sentence-BERT dense embeddings, and OpenAI CLIP multimodal representations in pgvector.
- **Cryospheric Sensors & Polar GIS Standards (Refs [08, 09]):** WMO Global Cryosphere Watch observation standards paired with Leaflet Antarctic/Arctic stereographic projections (EPSG:3031 / EPSG:3995).
- **Microservice Engineering & Containerization (Ref [10]):** FastAPI async REST endpoints, Pydantic v2 schemas, PostgreSQL/pgvector cluster, MinIO S3 object store, and Docker orchestration.

PROTOTYPE STATUS & EVALUATION ROADMAP

- **SIH26063 Prototype Platform:** POLARIS is presented as a functional prototype proposal for SIH 2026, fully addressing MoES / NCPOR problem specifications.
- **Standards-Backed Architecture:** Production stack (Next.js, FastAPI, PostgreSQL, pgvector, PyMuPDF, Leaflet, MinIO S3) ensuring feasibility and reproducibility.
- **Submission Demonstration & Verified Links:**
 - ▶ Video Demo: <https://youtu.be/HitV08Q82Fg>
 - ▶ GitHub Repo: <https://github.com/jayaram14329/polaris>
 - ▶ Live Prototype: <https://hopes-convertible-vcr-versions.trycloudflare.com>

SCAN FOR VIDEO DEMO



YouTube: HitV08Q82Fg
Click link or scan QR code